HARIRAMAKRRISHNAN RAMACHANDRAN

+353 89 970 6156 • hariramakrrish@gmail.com • Dublin, Ireland • **Stamp 1G — Full-Time Work Eligible**

PROFESSIONAL EXPERIENCE — 3.5 YEARS

Software Engineer — HCL Technologies, Chennai, India Sep 2021 – Jan 2025

- Built and optimized **Python** data pipelines on **AWS Lambda** and **S3** to ingest and process audio data, handling approximately **3,500 events/sec** during peak usage for model training.
- Developed and maintained **Java Spring Boot** REST APIs for backend services, implementing robust error handling and asynchronous processing patterns.
- Designed and implemented a relational database schema and optimized **SQL** queries in **PostgreSQL** for a large-scale data processing system, improving data retrieval efficiency by **30%**.
- Collaborated with **AI Scientists** to refine training datasets, ensuring data quality and format compatibility for next-generation model development.
- Discovered and integrated new sources of audio data, building reliable **Python** ingestion pipelines into our training datasets, expanding coverage by **15%**.
- Took ownership of engineering projects from problem discovery through implementation, including continuous improvement of large-scale data-processing systems for throughput and cost optimization.
- Containerized data processing services using **Docker** and deployed them to a managed **Kubernetes** cluster, configuring resource limits and enabling reproducible deployments.

SKILLS

Languages – Python, Java, Bash, SQL, R

Frameworks & Libraries – Spring Boot, pandas, NumPy, scikit-learn, FastAPI

Cloud & Infrastructure – AWS (EC2, S3, Lambda, CloudWatch), Docker, Kubernetes, GCP, Terraform

Data Processing & ETL – Data Pipelines, Large-Scale Data Processing, Data Acquisition, Web Crawlers

AI & Machine Learning – ML Training Data, Dataset Quality, Audio Processing, Model Training

Databases – PostgreSQL, MySQL, Oracle

Development Tools & Methodologies – Git, Jenkins, Jira, Agile, Linux

PROJECTS

Audio Data Ingestion Pipeline for ML Training

- Developed a scalable **Python** data pipeline on **AWS** to ingest diverse audio data sources, processing over **1 TB** of raw audio daily.
- Implemented data validation and cleaning routines using **pandas** to ensure high-quality datasets for downstream **AI model training**.
- Integrated with cloud storage solutions like **S3** and orchestrated processing jobs using **Bash** scripting for efficient data management.

Production Data Processing System Optimization

- Engineered and maintained a large-scale data-processing system using **Java Spring Boot** and **SQL**, enhancing throughput by **25%**.
- Optimized **PostgreSQL** database queries and implemented strategic indexing, reducing average query execution times by **40%** for analytical workloads.
- Containerized the application using **Docker** and managed deployments on a **Kubernetes** cluster, improving system reliability and scalability.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India 2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)